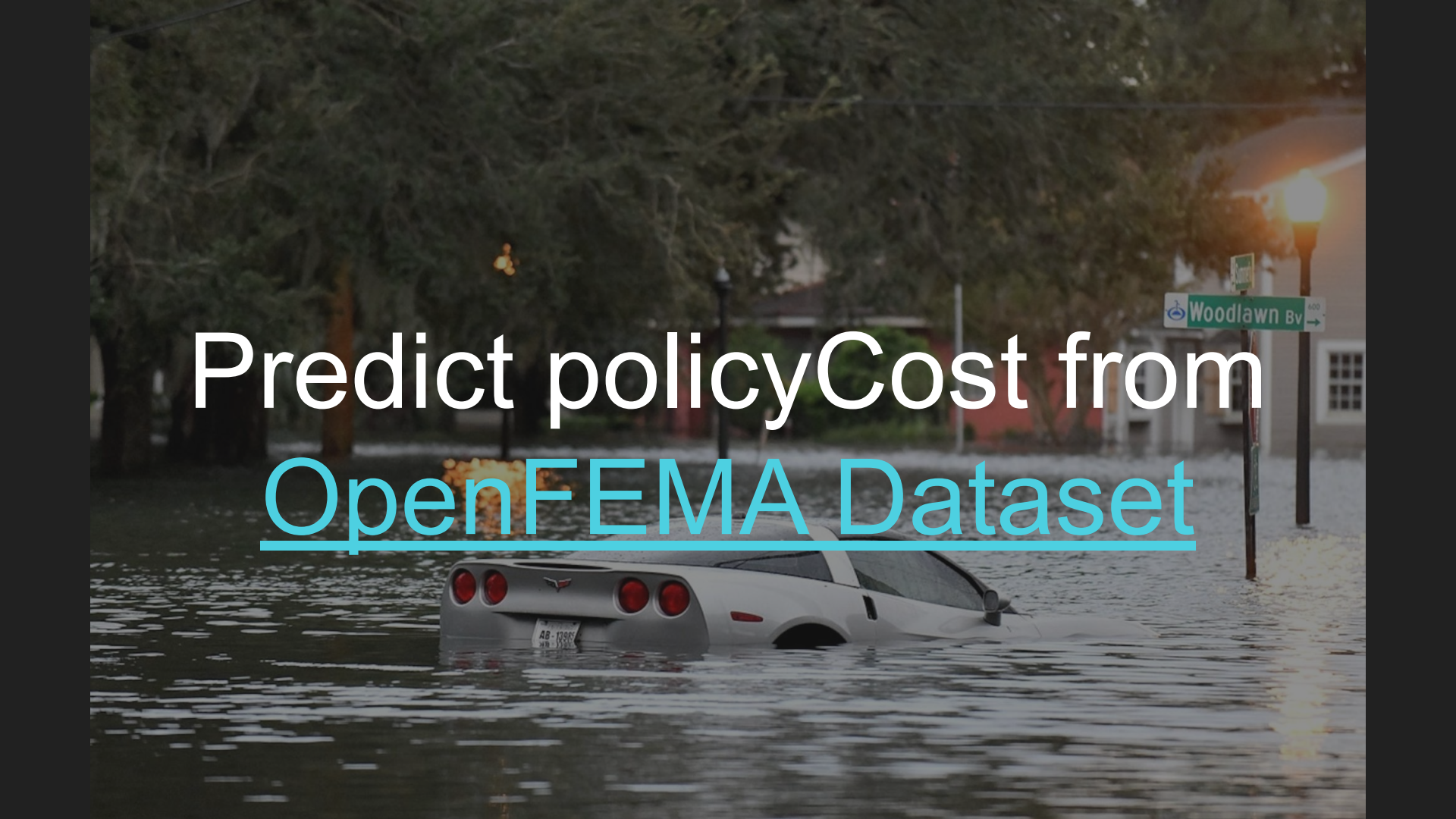
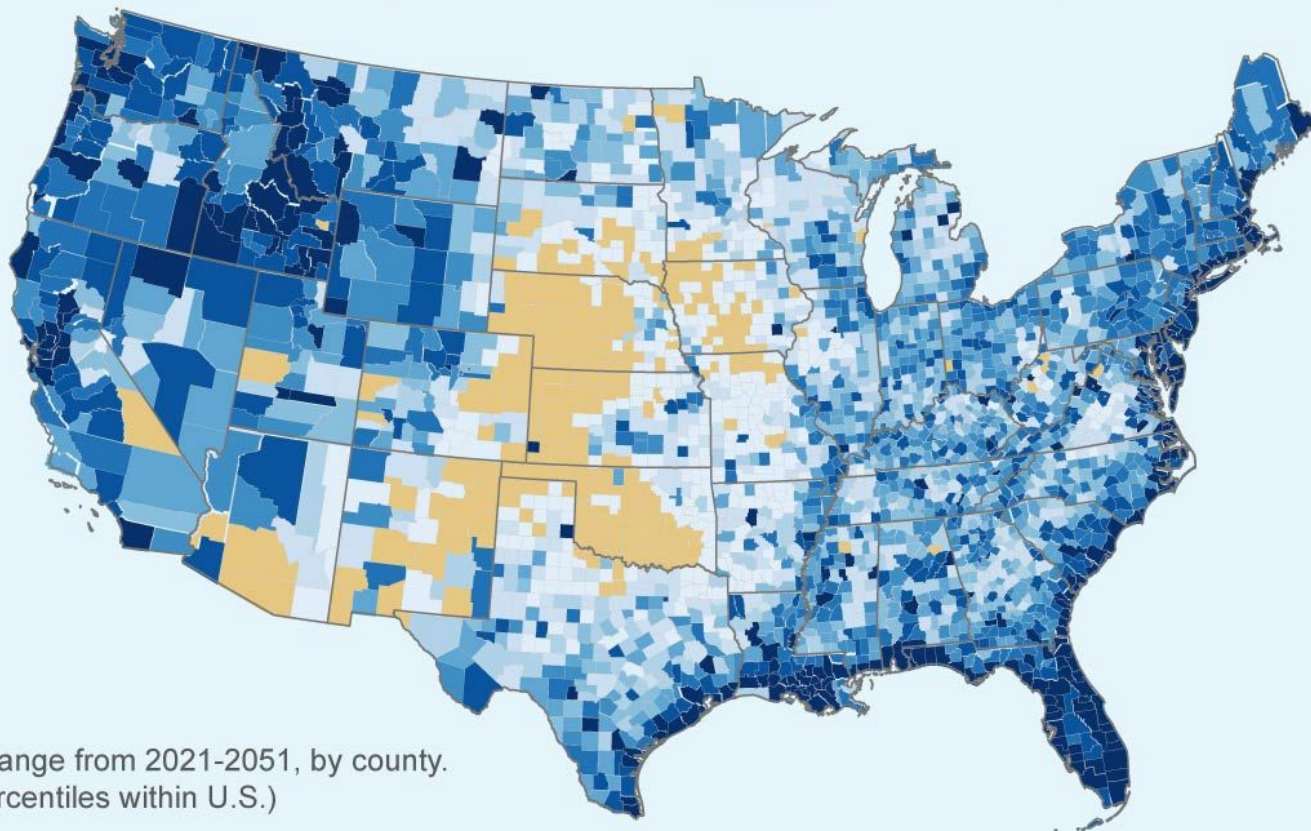




Predict policyCost from OpenFEMA Dataset

Change in flood risk over the next 30 years

Decreasing risk Increasing risk



Overall risk change from 2021-2051, by county.
(Change in percentiles within U.S.)

parquet

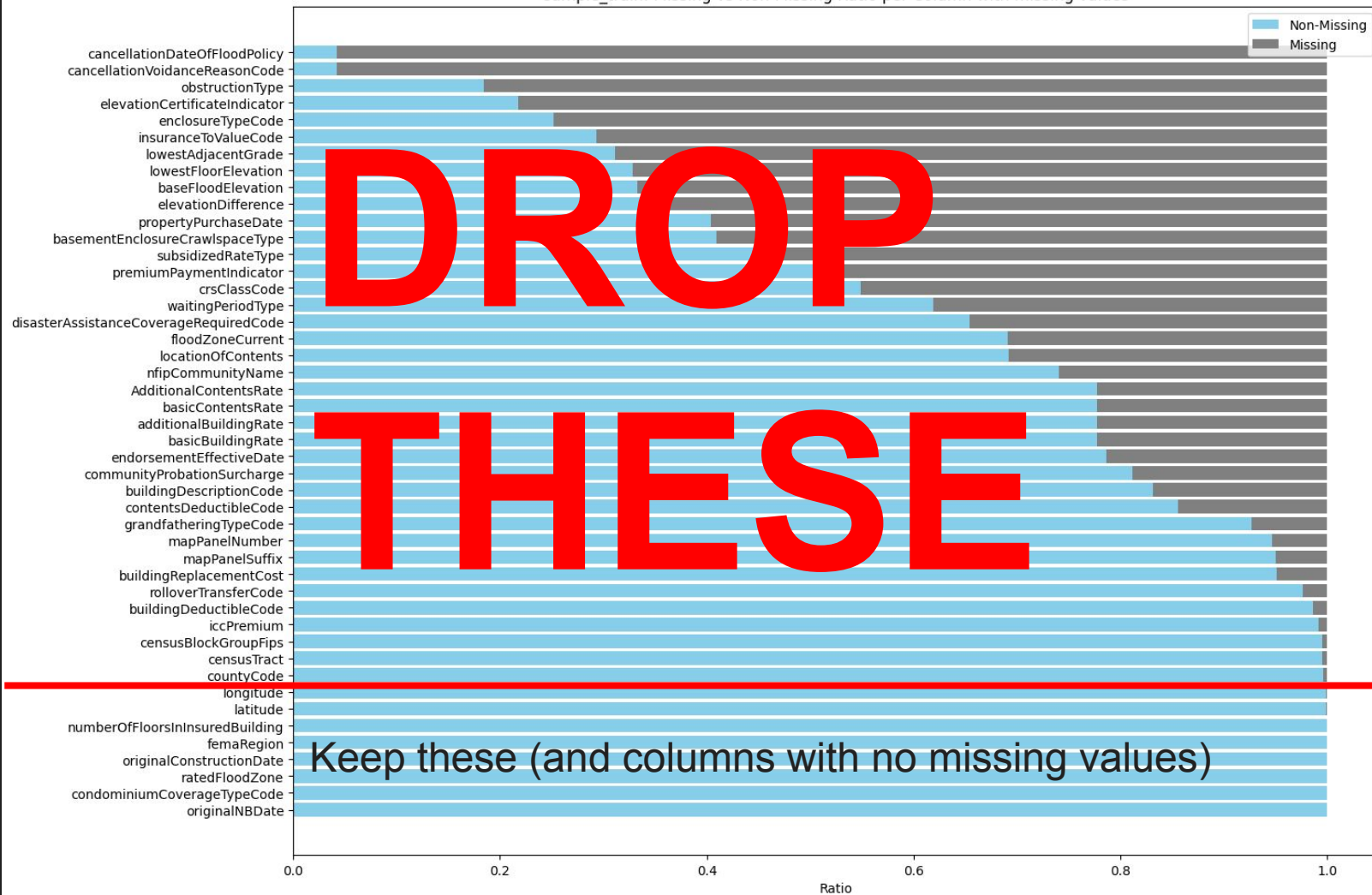
[Link to parquet](#)

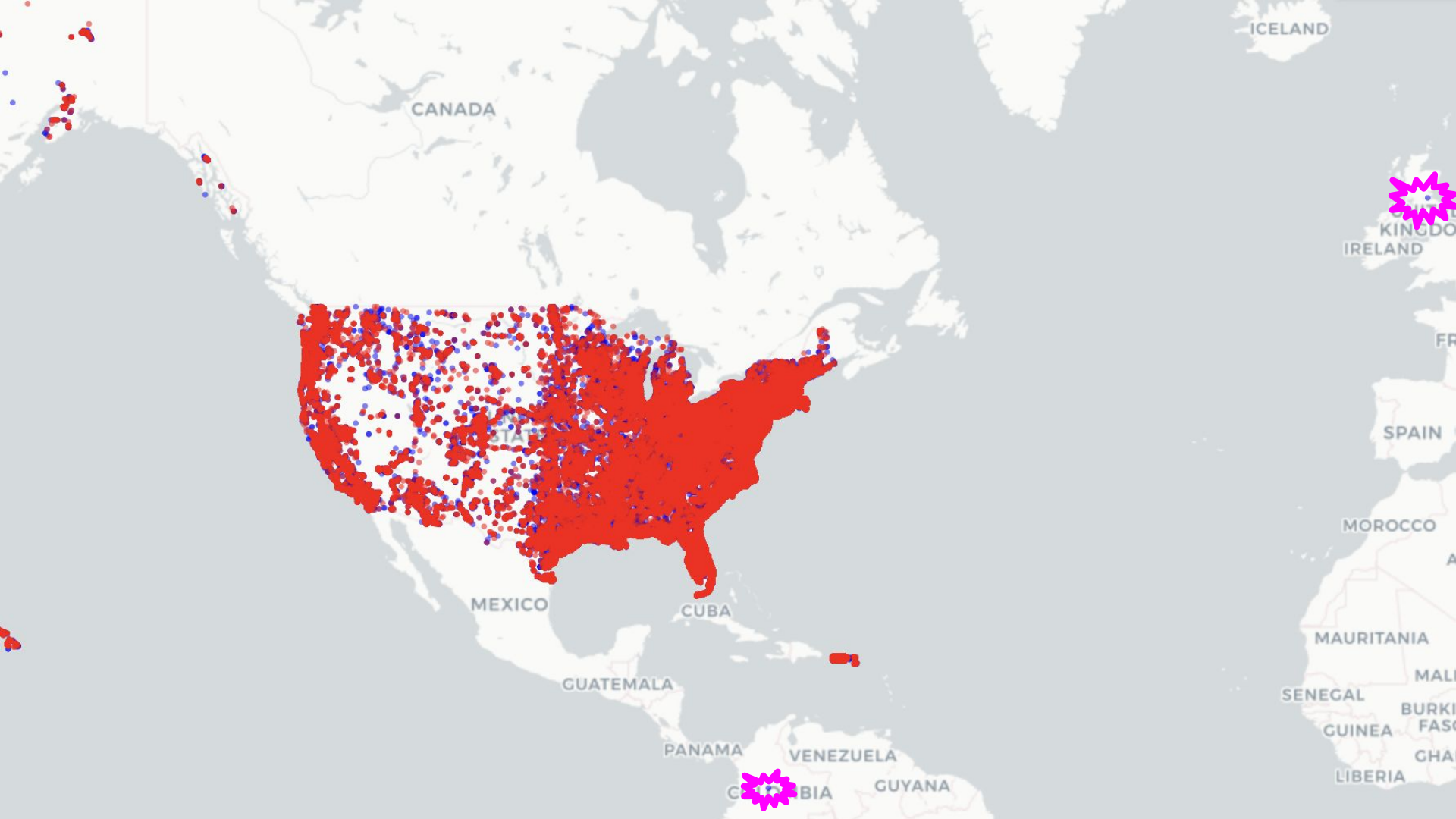
72583994

large (500MB - 10GB)

- Over **70 million** records!
- I sampled via hashing
 - ~350,000 for training
 - ~350,000 for testing

sample_train: Missing vs Non-Missing Ratio per Column with missing values





The points in Colombia and the UK are suspicious.

```
# Define the suspicious coordinates
coords_to_check = [(4.6, -74.1), (56.1, -3.2)]

# Filter the combined dataframe for these points
# We use a small epsilon for float comparison safety
epsilon = 0.1

mask = pd.Series(False, index=df_combined.index)
for lat, lon in coords_to_check:
    mask |= (df_combined['latitude'].sub(lat).abs() < epsilon) & (df_combined['longitude'].sub(lon).abs() < epsilon)

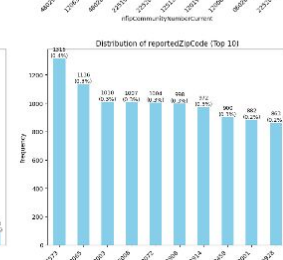
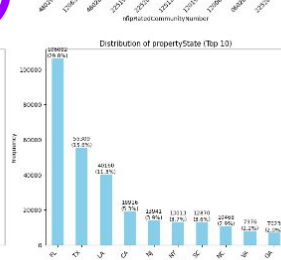
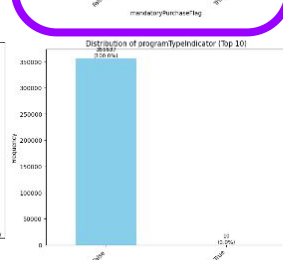
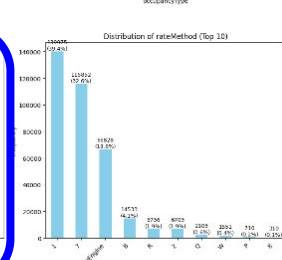
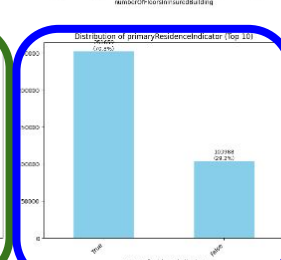
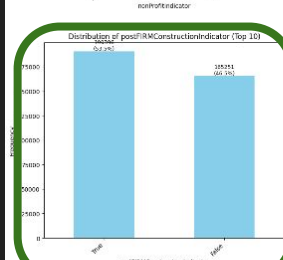
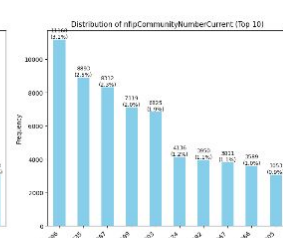
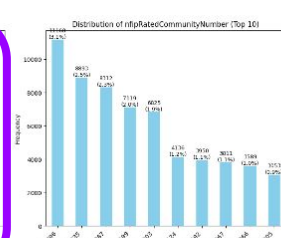
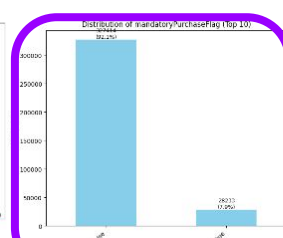
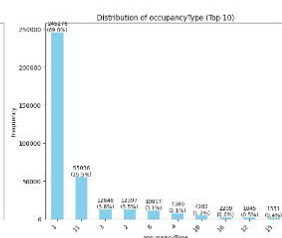
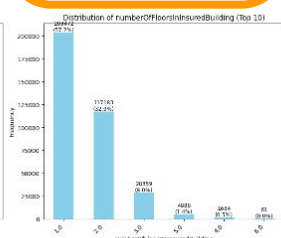
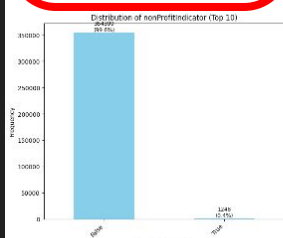
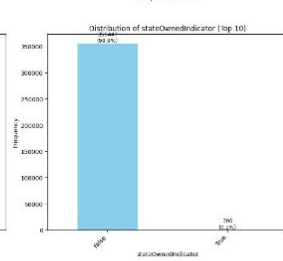
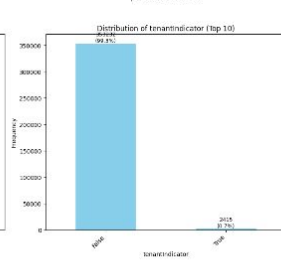
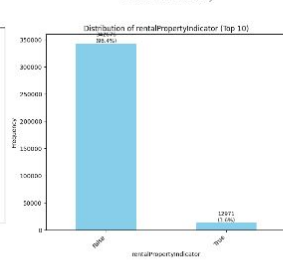
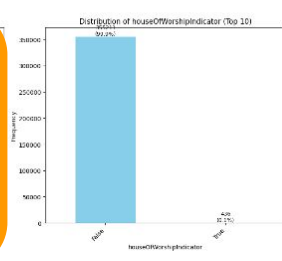
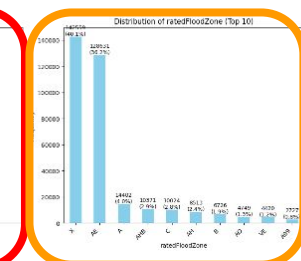
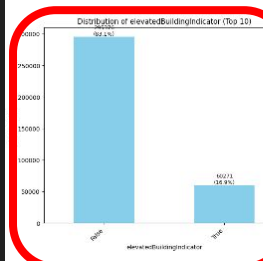
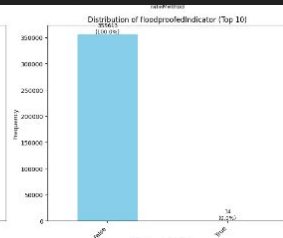
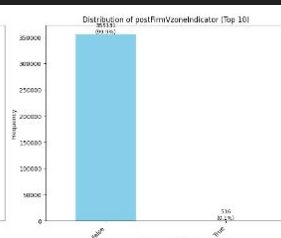
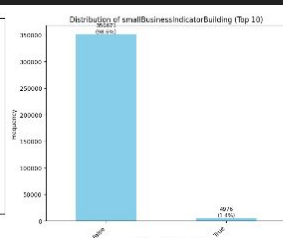
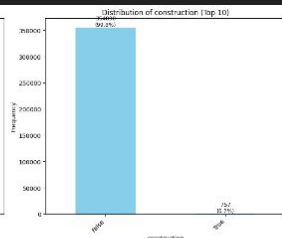
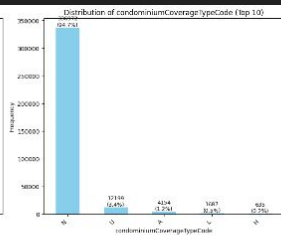
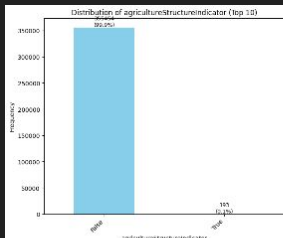
suspicious_rows = df_combined[mask]

print(f"Found {len(suspicious_rows)} rows at these coordinates:")
display(suspicious_rows[['propertyState', 'reportedZipCode', 'latitude', 'longitude']])
```

Found 2 rows at these coordinates:

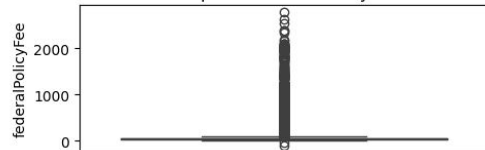
	propertyState	reportedZipCode	latitude	longitude
193990	MS	39759	56.1	-3.2
213838	PR	00949	4.6	-74.1

The latitude and longitude don't line up with state data. Let's drop these, along with all rows with missing values.

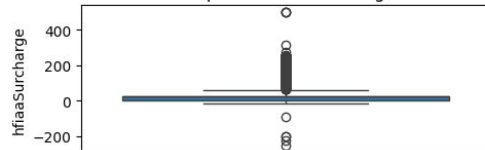


- Many features extremely unbalanced (over 90% the same value)
- To avoid curse of dimensionality, I only chose the most balanced
 - `elevatedBuildingIndicator`
 - `ratedFloodZone` (which I got dummies as X, AE)
 - `postFIRMConstructionIndicator`
 - `primaryResidenceIndicator`
 - `mandatoryPurchaseFlag`

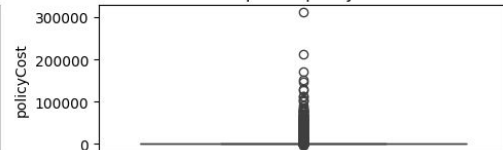
Boxplot of federalPolicyFee



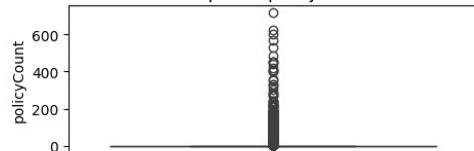
Boxplot of hfiaaSurcharge



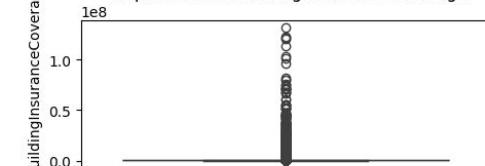
Boxplot of policyCost



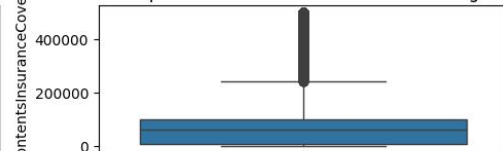
Boxplot of policyCount



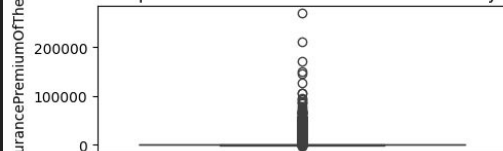
Boxplot of totalBuildingInsuranceCoverage



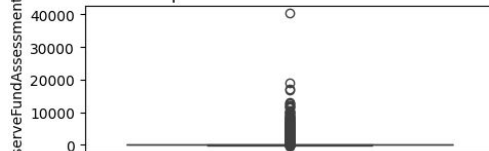
Boxplot of totalContentsInsuranceCoverage



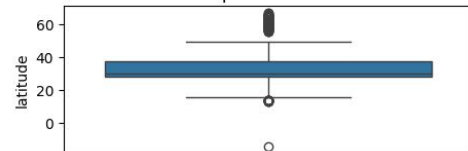
Boxplot of totalInsurancePremiumOfThePolicy



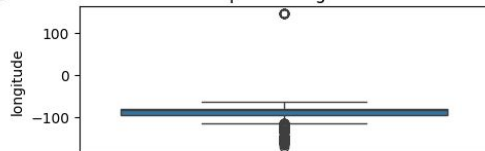
Boxplot of reserveFundAssessment



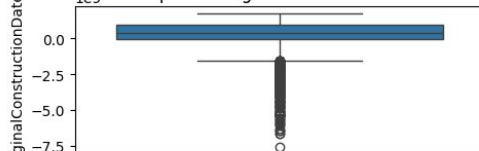
Boxplot of latitude



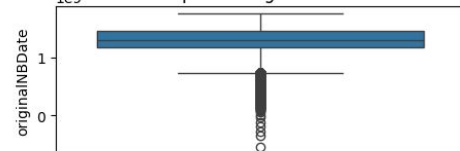
Boxplot of longitude



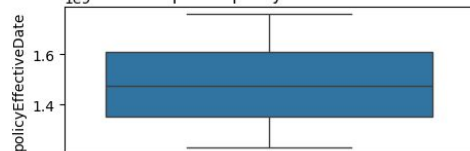
Boxplot of originalConstructionDate



Boxplot of originalNBDate



Boxplot of policyEffectiveDate



Boxplot of policyTerminationDate



Problems

1. Latitude, Longitude
2. Skew
3. Scale

Solution pipeline

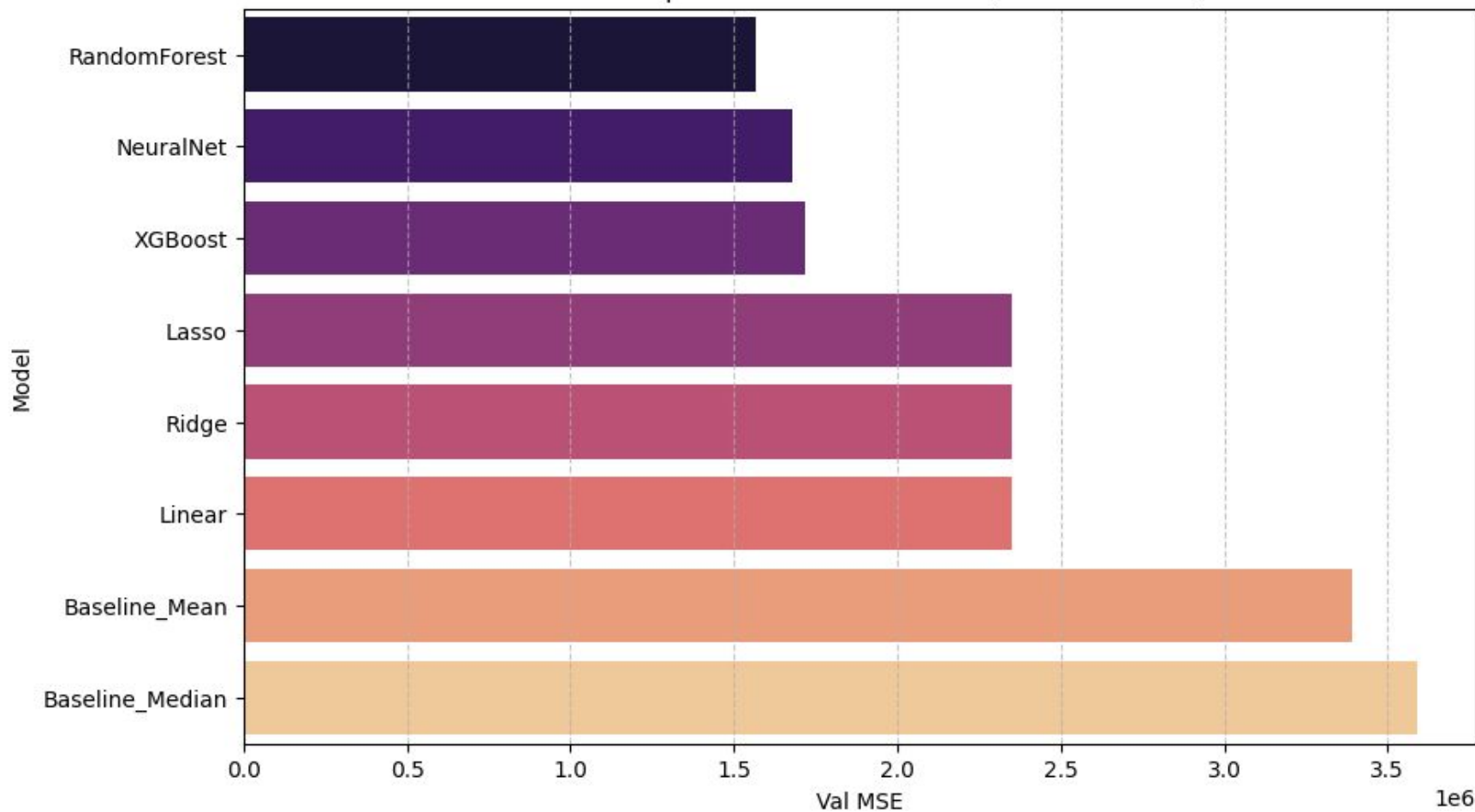
1. Cyclical encoding (3D Cartesian coords)
2. Reflect if left skew, log transform
3. Standard scaler

policyCost	Policy Cost	integer	Calculated in dollars by adding together calculated premium, reserve fund assessment, federal policy fee, and HFIAA surcharge
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Multicollinearity danger!

Model Comparison: Validation MSE (Lower is Better)



Winning Model: RandomForest

Test MSE

1779859.7177

Test RMSE

1334.1138

Test MAE

409.1102

Test R2

0.5058

Feature Importances (RandomForest)

